

half the power, extending laptop battery life. Additionally, the engine can handle multiple tasks simultaneously, like eight different tasks at the same time.

Speed and accuracy on one “Block”

AMD stressed the importance of having a powerful NPU that is ready for the AI age, focusing on how the NPU is far more efficient when it comes to performance per watt. This indicates how AMD’s XDNA 2 is all about high performance at the lowest possible power consumption. The XDNA2 NPU has also introduced “Block FP16,” which enables the processor to handle complex calculations faster and more accurately, without sacrificing precision. Normally, processors use either 16-bit floating point numbers (which are accurate but slow) or 8-bit integers (which are fast but less accurate). AMD claims Block FP16 combines the best of both, performing 8-bit integers with the accuracy of 16-bit floating point numbers.

Some sweet AMD AI laptops are about to arrive

Laptops on the AMD Ryzen AI 300 series processor will start hitting the shelves in the US market starting mid-July but there’s no word on their availability in India yet. However, we saw quite a few models from Asus, Acer, HP, and MSI being displayed at the demo zone. We can’t wait to get our hands on the machines for a more extended duration, for sure.

Our favourite pick of the lot has to be the HP OmniBook Ultra 14 and the Asus TUF A14. The HP Omnibook Ultra 14 is a typical thin and light laptop that is well-built and has a sharp 1440p display. The Asus TUF A14 is a thin and light gaming laptop with an NVIDIA GeForce RTX 4060 GPU paired with a 2.5K, 165Hz display. Both the laptops are powered by the AMD Ryzen AI 9 HX 370 processor, which is the highest in the line-up of AMD Ryzen AI 300 series processors. You can expect 150 AMD Ryzen AI 300 series-based laptops to hit the market in 2024, how many of them will make it to India, is anybody’s guess. Given that we’ve seen a fairly low



The Asus Zenbook S16 looked very impressive at the AMD Tech Day 2024 demo zone, the AMD Radeon 890M might just be the beginning of power-efficient entry-level gaming laptops. We await the review unit eagerly.

representation of AMD-based laptops in India in the recent past, we are hoping the trend will change this time around.

AMD Ryzen AI 300 Series: How Zen 5 Architecture Boosts CPU Performance

When AMD announced its Ryzen AI 300 series processors at Computex 2024, the focus was on AI performance and the XDNA 2 NPU. We’ve already covered the AMD XDNA 2 NPU architecture and how it achieves 50 TOPS of performance, which is the highest number so far as pure claimed NPU performance is concerned (Intel NPU 4 on Lunar Lake claims 48 TOPS).

AMD’s Zen 5 core, powering the upcoming Ryzen 9000-series CPUs and Ryzen AI 300-series processors, boasts a 16% average increase in instructions per clock (IPC) on the back of architectural enhancements and a node shrink to TSMC’s 4nm process technology. The Zen 5 architecture boasts significant improvements in performance and efficiency, thanks to a redesigned front end, wider execution pipelines, and enhanced cache hierarchy.

How is AMD Zen 5 getting those performance gains?

The new front-end changes include a revamped fetch, decode, and dispatch mechanism that yields more instructions

to the back end. Additionally, branch prediction accuracy has been improved, with more predictions per cycle. The dispatch and execution engine has been enhanced to support up to 8 instructions per cycle, with a unified scheduler and expanded execution window.

The cache hierarchy has also been improved, with a 50% increase in data cache size to 48 kilobytes, without increasing latency. Furthermore, the data bandwidth has been doubled, with support for 512-bit loads and stores. The AVX math unit has been doubled to a physical 512-bit wide pipeline, resulting in reduced latency and increased throughput.

Mark Papermaster, EVP and CTO at AMD, on Zen 5

“Zen 5 will not disappoint you with the kind of performance improvements that we’ve brought, and we won’t let up going forward,” said Mark Papermaster, EVP and CTO at AMD. “It’s going to be a pedestal that we’re going to build upon the next several generations of Zen,” he said. Mark Papermaster further emphasised, “We’re very happy with the improvements that we made. We could not be more proud with our new math acceleration unit.” Mark Papermaster reiterated their commitment to innovation, stating, “We will not slow down on our innovations at AMD.”

We’ll continue to deliver leadership and bring delightful experiences to our customers.” He added, “We think broadly with our IP designs in terms of how we deploy it, in terms of how we optimize it to be both high performant and highly efficient, and nothing has changed at all in that philosophy and that approach with Zen 5.” Zen 5 is poised to “absolutely excel at AI data preparation and general-purpose CPU workloads.” Mark Papermaster affirmed, “We’re maniacally focused on execution and delivering new capabilities on a regular cadence”, Mark affirms. **■**

